

Algorithm: Milk Collection, Profit Calculation, and Vaccination Reminder System

1. Start
2. Set VaccinationCount = 0
3. Read N (number of animals)
4. For Count = 1 to N:
 - a. Read MorningMilk (re-ask if negative)
 - b. Read EveningMilk (re-ask if negative)
 - c. TotalMilk = MorningMilk + EveningMilk
 - d. AmountEarned = TotalMilk * RatePerLiter
 - e. Display TotalMilk, AmountEarned
 - f. Read FeedingExpense (re-ask if negative)
 - g. Profit = AmountEarned - FeedingExpense
 - h. Display Profit
 - i. Read LastVaccinationDate
 - If (next vaccination due within 30 days):
Display "Vaccination Needed"
VaccinationCount++
 - Else:
Display "Vaccination Not Required"
 - j. End of loop
5. Display final report (totals, profit, VaccinationCount)
6. End

CODE

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#include <stdio.h>
#include <time.h>
#define RATE_PER_LITER 45.0 /* Rate per liter of milk */

int main() {
    int n, count, vaccinationCount = 0;
    char choice;

    printf("Enter number of animals: ");
    scanf("%d", &n);

    /* Arrays to store final report data */
    float totalMilkArr[n], earnedArr[n], expenseArr[n], profitArr[n];
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for (count = 1; count <= n; count++) {
    float morningMilk, eveningMilk, totalMilk, amountEarned, feedExpense, profit;
    int age;

    printf("\n--- Animal %d ---\n", count);

    /* ----- Morning Milk ----- */
    do {
        printf("Enter morning milk quantity (liters): ");
        scanf("%f", &morningMilk);

        if (morningMilk < 0) {
            printf("Invalid value. Skip this entry and set to 0? (y/n): ");
            scanf(" %c", &choice);
            if (choice == 'y' || choice == 'Y') {
                morningMilk = 0;
                break;
            }
        } else {
            break;
        }
    } while (1);

    /* ----- Evening Milk ----- */
    do {
        printf("Enter evening milk quantity (liters): ");
        scanf("%f", &eveningMilk);

        if (eveningMilk < 0) {
            printf("Invalid value. Skip this entry and set to 0? (y/n): ");
            scanf(" %c", &choice);
            if (choice == 'y' || choice == 'Y') {
                eveningMilk = 0;
                break;
            }
        } else {
            break;
        }
    } while (1);

    /* ----- Total Milk & Earnings ----- */
    totalMilk = morningMilk + eveningMilk;
    amountEarned = totalMilk * RATE_PER_LITER;

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printf("Total milk collected: %.2f liters\n", totalMilk);
printf("Amount earned: %.2f\n", amountEarned);

/* ----- Feeding Expense ----- */
do {
    printf("Enter feeding expense: ");
    scanf("%f", &feedExpense);

    if (feedExpense < 0) {
        printf("Invalid value. Skip this entry and set to 0? (y/n): ");
        scanf(" %c", &choice);
        if (choice == 'y' || choice == 'Y') {
            feedExpense = 0;
            break;
        }
    } else {
        break;
    }
} while (1);

/* ----- Profit ----- */
profit = amountEarned - feedExpense;
printf("Profit: %.2f\n", profit);

/* Store values for final report */
totalMilkArr[count - 1] = totalMilk;
earnedArr[count - 1] = amountEarned;
expenseArr[count - 1] = feedExpense;
profitArr[count - 1] = profit;

/* ----- Vaccination Check ----- */
int lastVaccDay, lastVaccMonth, lastVaccYear;

printf("Enter last vaccination date (DD MM YYYY): ");
scanf("%d %d %d", &lastVaccDay, &lastVaccMonth, &lastVaccYear);

// Create structure for last vaccination date
struct tm lastVacc = {0};
lastVacc.tm_mday = lastVaccDay;
lastVacc.tm_mon = lastVaccMonth - 1; // Months are 0-11
lastVacc.tm_year = lastVaccYear - 1900; // Years since 1900

// Convert to time_t

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time_t lastVaccTime = mktime(&lastVacc);

// Get current date and time
time_t currentTime = time(NULL);

// Calculate difference in seconds and convert to days
double diffSeconds = difftime(currentTime, lastVaccTime);
int daysSinceLastVacc = (int)(diffSeconds / (60 * 60 * 24));

printf("Days since last vaccination: %d\n", daysSinceLastVacc);

// Assuming vaccination interval is 30 days
if (daysSinceLastVacc >= 30) {
    printf("Vaccination Needed\n");
    vaccinationCount++;
} else {
    printf("Vaccination Not Required\n");
}

printf("Vaccination Count: %d\n", vaccinationCount);

}

/* ----- Final Report ----- */
printf("\n===== FINAL REPORT =====\n");
for (count = 1; count <= n; count++) {
    printf("\nAnimal %d:\n", count);
    printf(" Total Milk    : %.2f liters\n", totalMilkArr[count - 1]);
    printf(" Amount Earned   : %.2f\n", earnedArr[count - 1]);
    printf(" Feeding Expense: %.2f\n", expenseArr[count - 1]);
    printf(" Profit          : %.2f\n", profitArr[count - 1]);
}
printf("\nTotal animals requiring vaccination: %d\n", vaccinationCount);
printf("===== \n");

return 0;
}

```